

Chapter 5, Section 2

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1. If X is a birationally ruled surface, show that the curve C , such that X is birationally equivalent to $C \times \mathbb{P}^1$, is unique up to isomorphism.

Proof. It is enough to show $C \times \mathbb{P}^1$ and $C' \times \mathbb{P}^1$ are birational if and only if C is isomorphic to C' . $\square$

2. Let X be the ruled surface $\mathbb{P}(\mathcal{E})$ over a curve C . Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

Proof. Let $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C(\epsilon)$, with invariant $e = -\deg \epsilon \geq 0$. Defining a section $C \rightarrow X$ is equivalent to giving a surjection $\mathcal{E} \rightarrow \mathcal{L} \rightarrow 0$. The two natural choices for $\mathcal{L}$ are the direct summands $\mathcal{O}_C$ and $\mathcal{O}_C(\epsilon)$. The surjection $\mathcal{E} \rightarrow \mathcal{O}_C(\epsilon) \rightarrow 0$ correspond to the section $C_0 \subset X$ such that $\mathcal{O}_X(C_0) \cong \mathcal{O}_X(1)$. By (2.9), the section D corresponding to $\mathcal{E} \rightarrow \mathcal{O}_C \rightarrow 0$ has degree 0, which means $C_0 \cdot D = 0$.

Conversely, suppose we have two surjections $\mathcal{E} \rightarrow \mathcal{L}' \rightarrow 0, \mathcal{E} \rightarrow \mathcal{L}'' \rightarrow 0$ corresponding to C', C'' , respectively. Let $\mathcal{N}'$ and $\mathcal{N}''$ be the kernels.

$$\mathcal{N} = \mathcal{O}_C(\epsilon - \epsilon'), \quad \mathcal{N} = \mathcal{O}_C(\epsilon - \epsilon'')$$

$\square$

3. (a) If $\mathcal{E}$ is a locally free sheaf of rank r on a nonsingular curve C , then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i/\mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \dots, r$. We say that $\mathcal{E}$ is a *successive extension* of invertible sheaves.

- (b) Show that this is false for varieties of dimension ≥ 2 . In particular, the sheaf of differentials $\Omega_{\mathbb{P}^2}$ on $\mathbb{P}^2$ is not an extension of invertible sheaves.

4. Let C be a curve of genus g , and let X be the ruled surface $C \times \mathbb{P}^1$. We consider the question, for what integers $s \in \mathbb{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.

- (a) Show that $r = 0$ and $r \geq g + 1$ are always possible.
 (b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

5. *Values of e .* Let C be a curve of genus $g \geq 1$.

- (a) Show that for each $0 \leq e \leq 2g - 2$ there is a ruled surface X over C with invariant e , corresponding to an indecomposable $\mathcal{E}$.
 (b) Let $e < 0$, let D be any divisor of degree $d = -e$, and let $\xi \in H^1(X, \mathcal{O}_X(-D))$ be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_X(D) \longrightarrow 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $H^0(X, \mathcal{O}_X(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \not\subseteq H$.

- (c) Now show that if $-g \leq e < 0$, there exists a ruled surface X over C with invariant e .
 (d) For $g = 2$, show that $e \geq -2$ is also necessary for the existence of X .

6. Show that every locally free sheaf of finite rank on $\mathbb{P}^1$ is isomorphic to a direct sum of invertible sheaves.
 7. On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2 = 1$ form a one-dimensional algebraic family, parametrized by the points of the base curve C , and that no two are linearly equivalent.
 8. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$, $\mathcal{F} \neq \mathcal{E}$, $\mathcal{F} \neq 0$, we have

$$(\deg \mathcal{F})/\text{rank } \mathcal{F} > (\deg \mathcal{E})/\text{rank } \mathcal{E}$$

Replacing $>$ by $\geq$ defines *semistable*.

- (a) A decomposable $\mathcal{E}$ is never stable.
 (b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg \mathcal{E} < 0$ (respectively, ≥ 0).
 (c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not semistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leq 2g - 2$, (2) an element $\mathcal{L} \in \text{Pic } C$ of degree $-e$, and (3) a nonzero $\xi \in H^1(X, \mathcal{L}^{-1})$, determined up to a nonzero scalar multiple.
 9. Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbb{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, in which case $\deg Y = 2a$, $g(Y) = (a - 1)^2$, or $\deg Y$ is odd, say $2a + 1$, and $g(Y) = a^2 - a$.
 10. For any $n > e \geq 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$ given by (2.19). If $n \geq 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geq 4$, there exists a nonhyperelliptic curve which has a g_3^1 .
 11. Let X be a ruled surface over the curve C , defined by a normalized bundle $\mathcal{E}$, and let ϵ be the divisor on C for which $\mathcal{O}_C(\epsilon) \cong \bigwedge^2 \mathcal{E}$ (2.8.1). Let b be any divisor on C .
 (a) If $|b|$ and $|b + \epsilon|$ have no base points, and if b is nonspecial, then there is a section $D \sim C_0 + bf$, and $|D|$ has no points.
 (b) If b and $b + \epsilon$ are very ample on C , and for every point $P \in C$, we have $b - P$ and $b + \epsilon - P$ nonspecial, then $C_0 + bf$ is very ample.
 12. Let X be a ruled surface with invariant e over an elliptic curve C , and let b be a divisor on C .
 (a) If $\deg b \geq e + 2$, then there is a section $D \sim C_0 + bf$ such that $|D|$ has no base points.
 (b) The linear system $|C_0 + bf|$ is very ample if and only if $\deg b \geq e + 3$.
 13. For every $e \geq -1$ and $n \geq e + 3$, there is an elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbb{P}^4$.
 14. Let X be a ruled surface over a curve of genus g , with invariant $e < 0$, and assume that $\text{char } k = p > 0$ and $g \geq 2$.
 (a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0$, then either $a = 1$, $b \geq 0$, or $2 \leq a \leq p - 1$, $b \geq \frac{1}{2}ae$, or $a \geq p$, $b \geq \frac{1}{2}ae + 1 - g$.
 (b) If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g - 1))$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.
 15. *Funny behaviour in characteristic p .* Let C be the plane curve $x^3y + y^3z + z^3x = 0$ over a field k of characteristic 3.
 (a) Show that the action of the k -linear Frobenius morphism f on $H^1(C, \mathcal{O}_C)$ is identically 0.

- (b) Fix a point $P \in C$, and show that there is a nonzero $\xi \in H^1(C, \mathcal{O}_C(-P))$ such that $f^*\xi = 0$ in $H^1(C, \mathcal{O}_C(-3P))$.
- (c) Now let $\mathcal{E}$ be defined by ξ as an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0,$$

and let X be the corresponding ruled surface over C . Show that X contains a nonsingular curve $Y \equiv 3C_0 - 3f$, such that $\pi: Y \rightarrow C$ is purely inseparable. Show that the divisor $D = 2C_0$ satisfies the hypothesis of (2.21b), but is not ample.

- 16.** Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(X)$ are the same. This is false for a projective curve.
- 17.** (a) Let $\varphi: \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^3$ be the 3-uple embedding. Let $\mathcal{I}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ . Then $\mathcal{I}/\mathcal{I}^2$ is a locally free sheaf of rank 2 on C , so $\varphi^*(\mathcal{I}/\mathcal{I}^2)$ is a locally free sheaf of rank 2 on $\mathbb{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbb{Z}$. Determine l and m .
- (b) Repeat part (a) for the embedding $\varphi: \mathbb{P}^1 \rightarrow \mathbb{P}^3$ given by $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$, whose image is a nonsingular rational quartic curve.